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Effects of non-driving-related tasks on driving performance after takeover request in automated driving (Level 3): a simulator study

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Abstract

Drivers are more likely to engage in non-driving-related tasks (NDRTs) until a takeover request prompts them to resume vehicle control in Level 3 automated driving. The objective of the present study was to evaluate the impact of diverse categories of non-driving-related tasks on driving performance subsequent to a takeover request. The study was conducted using a truck and bus simulator. A total of 31 participants were involved in the study. Following the training, the drivers performed a driving session without any other activity, and then six different activities in three scenarios were tested. The six NDRTs examined included: working on a laptop, making a phone call, watching a video, taking an audio course, relaxation with closed eyes, and manual activity. The driving performance was evaluated using a variety of metrics, including takeover time, average speed, vehicle distance to the centre of the lane, and an assessment of situational awareness. The results indicated that all NDRTs impaired driving performance to a certain extent; however, the extent of influence differed for each type of activity. The riskiest activities identified were found to be working on a laptop, watching a video, and making a phone call, which aligns with theoretical assumptions. Statistically significant differences in takeover time and average speed were identified for relaxation, working on a laptop, performing a manual task, participating in an audio course and watching a video.

Keywords Automated driving, SAE level 3, Non-driving-related task, Takeover, Simulator study, Takeover time

1 Introduction

Road transport safety is a significant issue. Human factors have been determined to be the primary cause of more than 90% of accidents, thus the implementation of modern technology, including autonomous vehicles, is a potential means of minimising the risk of human error. The utilisation of autonomous technologies is poised to displace the conventional role of the driver, thereby mitigating incidents attributable to driver inattention and distraction [1, 2]. While it is improbable that full automation will replace human drivers entirely in the coming decades [3], a shift towards mixed traffic is expected.

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Autonomous vehicles are expected to share the road with manually driven cars. The introduction of autonomous technologies on a larger scale appears to be most realistic in the context of road freight transport. The integration of these technologies will yield several advantages, including the streamlining of the entire transport process and the resolution of issues pertaining to driver shortages. In autonomous driving mode, the drivers will have the opportunity to allocate their time to other activities. This prompts the question of suitable activities and their potential impact on road traffic safety in a broader context.

This paper evaluates the impact of non-driving-related tasks on driving performance after a takeover request. To this end, a study was conducted involving 31 participants who engaged with a truck and bus simulator. Six different activities in three scenarios were tested. The performance of the drivers is evaluated using various measures, including takeover time, longitudinal and lateral changes (average speed and distance of the vehicle to the centre of the lane) and a situational awareness assessment. The utilisation of these metrics facilitates the discernment of risky activities from their safe counterparts. The simulator method was supplemented with several other methods (e.g., SAGAT).

1.1 Level 3 (conditional automation) and takeover request
SAE International [4] has delineated six levels of automated driving: no automation (L0), vehicles with driver assistance (L1), partial driving automation (L2), conditional automation (L3), high automation (L4) and full automation (L5) [4]. The present study focuses on conditionally automated driving - Level 3.

The third level of automated driving enables the system to assume complete control of the steering in specified conditions [4]. Vehicles at L3 are capable of independent autonomous driving without the driver's participation. However, it is important to note that there may be situations in which the system is unable to handle these tasks independently. In such cases, a takeover request (TOR) is initiated, which involves a transition from autonomous to manual steering modes. This request is made to the driver, who is then responsible for taking control of the vehicle from the autonomous mode. This transition is

referred to as takeover performance or fallback performance (see Fig. 1). In summary, the drivers of L3 vehicles are not required to maintain manual control of the steering wheel or to monitor the surrounding traffic. While not actively driving, they are required to maintain situational awareness to promptly take over steering responsibilities, assess the prevailing traffic conditions, and react accordingly.

A takeover process is comprised of multiple stages. The initial stage is the perception of the takeover request, which can be facilitated through visual, auditory, and/or vibrotactile feedback. This is followed by a cognitive processing of the information. The third stage is the selection of a response or the making of an appropriate decision. The fourth stage involves the repositioning of the hands and feet on the steering wheel and pedals, thus re-establishing control. The fifth and final stage is the actual reaction, which may take the form of steering and braking with sufficient force [6–10].

1.2 Effects of non-driving-related tasks on takeover performance

Non-driving-related-tasks (NDRTs), which include activities such as listening to music, reading and texting, are categorised as tertiary tasks. A substantial body of research has indicated heightened levels of engagement in non-driving-related tasks during autonomous driving mode [11, 12]. The impact of non-driving activities on performance during takeover is examined (e.g. [13, 14]), and the effect of extended autonomous driving on the driver's ability to take over is studied (e.g. [15]). The extant literature reports an increase in workload caused by non-driving activities and a decrease in situational awareness. Thus, drivers may need a longer time to reorient themselves to the driving task [11].

In the event of a non-driving-related task interfering with the takeover, the resulting impact on the speed and quality thereof is significant. A deterioration of all monitored parameters is usually observed, including an increase in reaction time, a decrease in time-to-collision (TTC), and an increase in the number of collisions [16, 17]. Conversely, Shahini & Zahabi [8] found that driving without non-driving-related tasks leads to better performance, specifically in terms of shorter hands-on-wheel

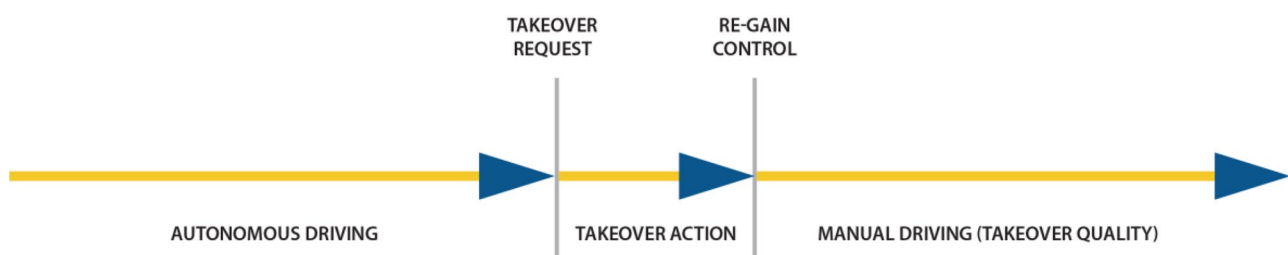


Fig. 1 Takeover process, cit. accord. To Lee et al. [5]

time, shorter braking time, longer TTC, and fewer accidents compared to driving with a non-driving activity. Moreover, drivers who performed non-driving related tasks exhibited a significantly higher takeover time compared to those who did not engage in such activities [18–21]. In addition, these drivers demonstrated a decline in driving performance [22, 23].

The assessment of non-driving-related tasks is typically based on characteristics that have the potential to impede the driver's capacity to execute the requisite steps necessary for the transition (e.g., recording a signal to take over, interrupting a non-driving activity, etc.). The demands that NDRTs place on the driver correspond to specific limitations in the ability to take control, based on which the individual dimensions of these activities can be defined [24]. A particularly salient feature pertains to the sensory modality of the activity. Numerous studies have demonstrated that the sensory modality of an activity exerts a substantial influence on the capacity to regain control [25–27]. The primary distinctions can be delineated between auditory and visual activities – auditory activities exhibit a diminished negative impact on driving performance when re-establishing control in comparison to visual activities. Furthermore, Zhang et al. [9] demonstrated that engaging in visual-auditory and visual-motor activities increased the time required to regain control, while no effect was observed with auditory-cognitive activities. The influence of the modality is increased by the further characteristics. Visual activities, when accompanied by a higher mental load, have been observed to result in the most significant deterioration in driving performance [13, 25–28]. Concentrating on the concept of visual attention has been shown to be advantageous [29]. The combination of visual modality and manual load while holding an object in the hand also significantly impairs driving performance during a takeover [25–27, 30, 31]. In the context of auditory activities, extant research has demonstrated that activities necessitating speech input or auditory resources result in enhanced driving performance with respect to lateral and longitudinal control of the vehicle, in addition to sufficient visual evaluation of the road when compared to visually and manually demanding tasks [26, 27].

Furthermore, research findings indicate that the duration of takeover time is increased when drivers engage in activities involving handheld devices or objects. The act of holding an object and then releasing it has been demonstrated to result in a prolonged duration required to regain control [9, 31, 32]. This has been shown to lead to an augmentation in reaction time, for instance, when individuals place their hands on the steering wheel [26, 33]. Conversely, the impact of operating integrated electronic devices (i.e. without holding them) on the process of taking control has been examined. The findings of this

study suggest that the operation of integrated electronic devices does not exert a substantial negative impact on performance [8]. However, concurrently, an escalating level of mental and visual load has been demonstrated to result in a decline in takeover performance [9].

Another significant criterion for the classification of activities is the degree of cognitive or visual load (demands of the activity). A substantial body of research has demonstrated that all types of loads have the capacity to exert a detrimental effect on an individual's ability to take over the wheel, thereby impacting driving performance (see [28, 32]). Visual load has been identified as a particularly salient factor in this regard. The elevated level of load has been demonstrated to result in an escalation in the driver's mental load [13, 20, 32, 34], which has been shown to precipitate a decline in performance during the takeover, particularly in circumstances where the prevailing traffic conditions are characterised by heightened demands and complexity [31].

When the driver turns his attention to NDRTs, his situational awareness decreases because his limited attentional resources are no longer used to monitor the car and the overall traffic situation [12]. If the activities are more demanding, the increased mental load prevents the driver from developing sufficient situational awareness, which secondarily leads to a poor assessment of the situation and erroneous reaction [23, 35, 36].

2 Method

2.1 Participants

A total of 35 participants – professional drivers took part in the experiment on the truck driving simulator from September to December 2023. The final sample for the purposes of all subsequent analyses comprised 28 male and 3 female participants. The data of 4 participants was excluded due to symptoms of simulator sickness. The mean age of the participants was 40 years (range: 23–57 years). The plurality of participants reported between 11 and 20 years of experience (32.3%, $n = 10$). Furthermore, the plurality of participants (35.5%; $N = 11$) drove between 40,001 and 60,000 km per year.

2.2 Ethical approval

The ethical process was ensured through a code of ethics, which is guaranteed by the ethics committee of the research institution CDV – Transport Research Centre (Líšeňská 33a, 636 00 Brno, Czechia). The study was approved by this committee. The participants were informed about the potential risks associated with the experiment and were given the option to withdraw from the study at any time. Informed consent was obtained from all participants prior to the commencement of the experiment. The experiment was overseen by an

instructor to ensure the safety of the participants. Participants were financially rewarded.

2.3 Apparatus

The primary measurement was conducted on a truck and bus simulator. The system is a complex one, consisting of several elements: an original Volvo truck cab (which was adapted for performing NDRTs), a motion platform simulating physical forces during vehicle driving, a visual system representing the driver's field of vision, an operator centre, a PC, and SimWorld Software. This high-fidelity moving base simulator represents the road and traffic environment, which is projected on screens in front of the vehicle with a broad field of view. Two screens positioned behind the vehicle display the scenery for the rearview mirrors. The driving performance following the takeover is recorded by the simulator software, including parameters such as the takeover time, speed, braking, and distance to the centreline (See Fig. 2).

2.4 Questionnaires

All participants completed questionnaires concerning their demographics, driving history, and frequency of performing activities in real life. A range of additional methods were employed, including the SAGAT method (Situational Awareness Global Assessment Technique) [23, 37], the NASA-TLX questionnaire (mental workload), and the Acceptance questionnaire of AV [38]. The participants were observed during their driving, and the view into the vehicle cabin was recorded. Following the conclusion of the experiment, the participants were invited to evaluate the perceived impact of the activities on takeover performance.



Fig. 2 Truck driving simulator with moving base

2.5 Experimental design, driving scenario and TOR

2.5.1 Experimental design

Each participant attended three sessions. The first session was an introduction to the vehicle control practice and instructions for taking over control was given, with a duration of five minutes. The second session involved a non-task session (baseline), during which the driving situation was observed in autonomous mode, with the session always concluding with the participant taking control and performing manoeuvres A, B, and C through three types of scenarios. The third session was a tasks session (experimental driving), during which six non-driving-related tasks were performed during autonomous driving and the participant took over driving. The duration of this session was 60 minutes.

2.5.2 Scenario

The autonomous vehicle was travelling on the motorway (favourable weather conditions and medium traffic density) in the right lane at a speed of 80 km/h. The driver was engaged in a non-driving-related task when an unexpected notification was received, prompting the initiation of manual driving. The signal was consistently generated approximately 10 seconds prior to the occurrence described below, i.e., approximately 220 metres before the driver was required to take control of the vehicle at the latest. The experiment is illustrated graphically in Fig. 3 below.

2.5.3 Takeover request (TOR)

The drivers were informed that the primary objective was to execute a safe driving manoeuvre, with the speed of the reaction being of secondary importance, provided it occurred within 10 seconds. Participants were provided with a set of guidelines to facilitate the desired response. These guidelines are outlined below: 1. Sound the signal (maximum duration of 10 seconds), 2. Orient oneself within the situation, 3. Announce the takeover by pressing the horn, 4. Drive manually. Participants declared taking control by pressing the takeover button (on the steering wheel), which switched the vehicle from autonomous mode to manual vehicle control.

The takeover scenario consisted of three types of non-emergency situations. The situation leading to the need to take over manual control was defined so that one of three manoeuvres was correct during manual control: A - continue smoothly in the right narrowed lane, B - avoid the obstacle in the right lane, i.e. turn into the left lane, while not endangering or restricting the vehicles driving there, C - safely bring the vehicle to a complete stop in the right-hand lane and create a rescue lane for the convoy of vehicles.

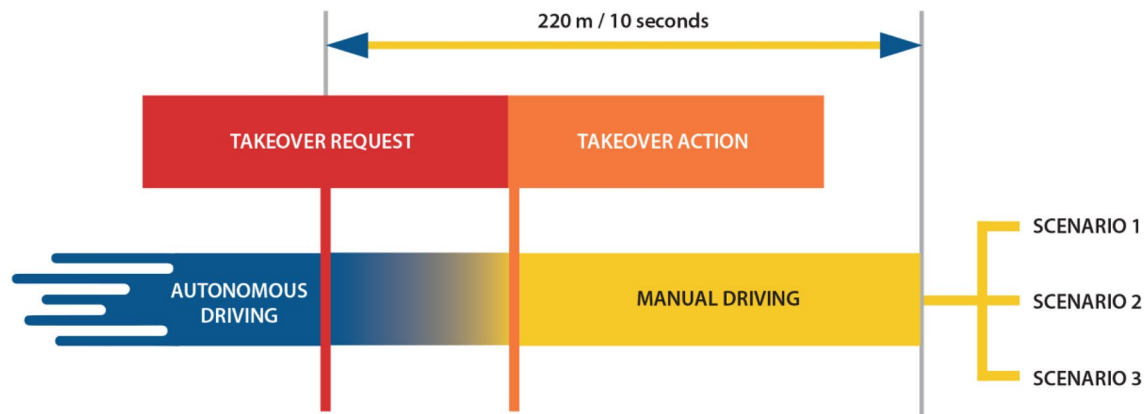


Fig. 3 Experimental design

Table 1 Description and characteristics of non-driving related tasks

NDRT	Description	Type of sensory modality	Mental load	Holding an object in hand
<i>Working on a laptop</i>	reading text, answering questions	visual	high	yes
<i>Making a phone call</i>	professional market research phone call	auditory	medium	no
<i>Watching a video</i>	educational video	visual	medium	no
<i>Listening to an audio course</i>	professional course with questions	auditory	medium	no
<i>Relaxation with closed eyes</i>	guided relaxation with music	auditory	low	no
<i>Manual task</i>	assembling Lego cubes	visual	low	yes

2.6 Non-driving-related tasks

Each of the six non-driving-related tasks was performed for a duration of 10 minutes, thereby ensuring that the driver was adequately immersed in the activity (see also [5, 20, 39]. Thereafter, the takeover signal was obtained. The instruction to perform the activity to the best of their ability was given, since their performance would be evaluated (working on a laptop) or because they would be asked about the presented content (watching a video and audio course). The sequence and concise description of the tasks are outlined in Table 1:

2.7 Data measurement and processing

2.7.1 Objective measures

To ascertain the driver's takeover performance, the following parameters were recorded: takeover time, braking and acceleration, speed, and position in the lane. The term "takeover time" refers to the interval that begins when the signal is received and ends when the button to engage manual vehicle control is pressed. Lateral control was measured by determining the position of the vehicle in relation to the lane's centre line. This variable describes the distance to the centre of the lane of the vehicle during the takeover manoeuvre. The standard deviation of the vehicle's lateral position (SDLP) is a related variable. The SDLP has been shown to provide insight into a driver's ability to maintain the position during the manoeuvre [40]. In the context of longitudinal control, the focus shifted to the vehicle's speed. It is hypothesised that smaller deviations in lateral position and longitudinal

speed indicate more consistent and seamless control [10]. All parameters were measured in both non-task sessions (baseline) and task sessions (with the influence of activity).

2.7.2 Self-reported measures

Several self-reported measures were evaluated. Situational awareness, measured by the SAGAT method (Situational Awareness Global Assessment Technique) [23, 37], was identified as the most significant factor. Following each takeover, the ride was halted, and the participant was debriefed about any objects, pedestrians, or traffic signs they had observed in the scene. This process enabled the determination of the activity's impact on situational awareness and its subsequent deterioration.

2.8 Data analysis

Each participant provided data from self-reported questionnaires and experimental data from the simulator drive (baseline without NDRT and six NDRTs). As three different scenarios were tested, we need to compare the results that are aligned to the same situations. The second objective was to compare the influence of the activities with each other. The statistical analysis was conducted using SPSS 24, with descriptive statistics, means and the Wilcoxon Signed-Rank test, a non-parametric alternative to the paired t-test, employed at a significance level of 0.05.

Table 2 Mean takeover time

Task	Mean	SD	p
Relaxation	4.00	1.39	0.001
Making a phone call	3.06	1.21	0.545
Driving without activity1	2.97	0.91	
Audio course •	2.97	1.25	0.057
Working on laptop •	4.26	1.99	0.000
Driving without activity2•	2.58	0.92	
Manual task ••	4.81	1.94	0.000
Watching a video ••	3.71	1.87	0.004
Driving without activity3••	2.61	1.48	

Legend: Paired activities are symbol-coded (no•, •, ••) to baseline (according to the same scenarios)

Table 3 Average speed during driving after takeover

Task	Mean	SD	p
Relaxation	30.8	12.2	0.002
Working on a laptop	35.6	11.4	0.000
Driving without activity1	51.9	10.9	
Audio course •	44.7	8.0	0.000
Making a phone call•	49.7	7.7	0.920
Driving without activity 2•	49.9	7.6	
Manual task••	27.4	9.4	0.002
Watching a video••	29.3	7.5	0.009
Driving without activity 3••	34.4	8.8	

Legend: Paired activities are symbol-coded (no•, •, ••) to baseline (according to the same scenarios)

3 Results

3.1 Takeover time

The measurement of the takeover time was determined by the interval between the signal for the takeover being sounded and the pressing of the takeover button. The average time required to regain control without performing the activity was 2.8seconds ($M_1=2.97$; $SD=0.91$; $M_2=2.58$; $SD=0.92$; $M_3=2.61$; $SD=1.48$). For activities, the average time taken was 3.8seconds. The average times for these activities are presented in Table 2. For each group of activities leading to the same scenario (marked with the same symbol), the data for driving with two types of activities and driving without activity (baseline) are always shown. The shortest recorded takeover time was for the audio course ($M=2.97$; $SD=1.25$), while the longest was for the manual task ($M=4.81$; $SD=1.94$). Where significant, the difference between the baseline and task session takeover times can be attributed to the influence of activities, namely relaxation ($p=0.001$), working on a laptop ($p=0.000$), manual tasks (0.000), and watching a video (0.004). Conversely, activities such as making a phone call ($p=0.545$) and participating in an audio course ($p=0.057$) did not result in a significant prolongation of the average takeover time.

Table 4 Distance to the centre lane during driving after taking control

Task	Mean	SD	p
Relaxation	0.13	0.25	0.124
Working on laptop	0.17	0.14	0.440
Driving without activity 1	0.14	0.25	
Audio course•	0.06	0.20	0.741
Making a phone call•	0.16	0.27	0.778
Driving without activity 2•	0.14	0.23	
Manual task••	0.10	0.18	0.605
Watching a video••	0.10	0.20	0.520
Driving without activity 3••	0.08	0.17	

Legend: Paired activities are symbol-coded (no•, •, ••) to baseline (according to the same scenarios)

Table 5 Situational awareness

NDRT	Mean (%)
Making a phone call	58.08
Manual activity	72.60
Working on laptop	74.18
Audio course	83.87
Relaxation	97.80
Watching video	98.38

3.2 Lateral and longitudinal changes in takeover performance

The average speed was calculated within a time interval of 10seconds after the takeover. The results for individual activities within the same scenario are presented in Table 3. For each activity group leading to the same scenario (same symbol), data for driving with two activity types and without activity (baseline) are shown. Statistically significant differences between the average speed in the baseline and during driving while performing an activity were demonstrated for several activities, namely relaxation ($p=0.002$), working on a laptop ($p=0.000$), audio course ($p=0.000$), manual task ($p=0.002$), and watching a video ($p=0.009$).

Table 4 contains data on the distance of the vehicle from the centre of the lane during driving after taking control. Each scenario group (same symbol) includes data for driving with two activity types and a driving without activity (baseline). The mean values and standard deviations (a measure of variability) were considered. Statistically significant differences in distance from the centre of the lane between the baseline and driving while performing an activity were not demonstrated for any activity.

3.3 Self-reported measures

As illustrated in Table 5, the results of the situational awareness battery are demonstrated in the form of the mean proportion of accurate responses to each sub-question. The results have been arranged in ascending order. It was observed that the participants demonstrated the least attention to their environment during the act

of making a phone call. Conversely, the highest level of overview was demonstrated by participants when watching a video.

4 Discussion

The present study investigated the influence of NDRTs on takeover performance following a TOR. The findings indicated that all forms of NDRTs adversely affected driving performance, though the extent of this impact varied.

The experimental results demonstrated that variations in reaction time can be attributed to the impact of the activity. The range of takeover time was from 2.53 to 2.97 seconds in the absence of any tasks, and from 3.06 to 4.81 seconds when tasks were performed. Statistically significant differences were identified for various activities, including relaxation, working on a laptop, and watching a video. The findings are consistent with those of other studies. Eriksson & Stanton [18] also refer to an average takeover time of 2.97 seconds. Petermann-Stock et al. [41] report a takeover time of 3.2 seconds. A meta-analysis by Soares et al. [42] of 36 studies revealed a range of reaction times between 1.5 and 3.5 seconds, based on various experimental conditions. Müller [40] examined takeover time as the duration from the presentation of the TOR to the application of a steering angle of more than 3° or pressing the brake by more than 10%. This time ranged from 1.10 to 1.64 seconds, with the longest times observed for watching videos, texting, and reading. The findings of this study corroborate the notion that tasks characterised by high levels of cognitive load and those involving visual stimuli can harm the temporal dynamics of the aforementioned processes.

The present study posits that an elongated takeover time, in conjunction with variations in the activities, signifies a diminished driver response, attributable to the attributes of the activity in question, including manual, cognitive, or visual demands. These characteristics contribute to an increased latency, especially when compared to the baseline. In summary, manual tasks, laptop work, and video watching have a detrimental effect on the takeover time. Laptop work is an activity with high visual and mental load, and the computer is handheld, which causes deterioration of driving performance [9, 13, 28]. This activity requires significant cognitive and visual attention, making it harder for participants to shift focus and respond quickly to the signal. This activity likely imposes a considerable cognitive load, thereby reducing reaction speed. Furthermore, Wandtner et al. [26] posit that watching a video is a visual activity with a negative effect. The results of the study demonstrated a performance impairment related to the holding of an object in hand, as indicated by increased reaction time [9, 31, 32]. Manual activities also resulted in longer takeover times [9, 31, 32]. The influence in the presented

study is considered high, with the possibility of impairment by fatigue given that it was the final activity. The need for physical manipulation during the task is likely to distract from the primary task of driving, making it difficult for participants to react promptly. The telephone call exhibited a medium level of influence on the duration of the takeover. Notwithstanding the possibility of cognitive distraction, phone calls enabled a rapid takeover. This phenomenon may be attributable to various factors, including the participants' optimal preparation at the commencement of the test, given that phone calls constituted the second activity, or the presence of unoccupied hands during the task. The relatively swift reaction observed indicates that, while phone calls do induce a degree of distraction, they do not significantly impede the ability to regain control within the temporal context of the task. The favourable outcome may be attributed to the practice effect associated with performing this activity in everyday life. The shortest observed takeover time was during relaxation. The enhanced responsiveness is likely attributable to the unobstructed hands and the restful state. This finding suggests that relaxation activities may facilitate a heightened state of readiness to regain control. Under the findings of other studies [26, 28], it is concluded that activities devoid of visual components (e.g., phone call, relaxation, audio course) result in more efficient takeover time in comparison to activities incorporating visual components.

It is important to note that a quick reaction (i.e., a short takeover time) does not necessarily indicate a high-quality response. To assess the quality of the response, qualitative parameters such as driving speed must be taken into consideration. Speed is a critical factor that reflects safe versus risky behaviour after taking control of the vehicle. It is generally accepted that driving at a reduced speed and maintaining adequate braking are indicative of a safer driving style. Conversely, behaviours associated with increased speed and diminished braking are commonly linked to a less safe takeover. The speed of passing through situations is partially related to the type of scenario that the participants dealt with. To this end, the baseline of driving without activity is also provided, and the results are compared as paired samples. The study found that there were usually similar trends in speed changes between pairs of activities within the same scenarios. It is evident that participants consistently adopt a slower pace after engaging in the activity, a phenomenon that can be attributed to the tendency to decelerate to achieve an adequate level of situational awareness. The distance from the centre lane has been utilised in numerous studies (see [28, 40]). It is widely accepted that greater variability in the distance from the centre line generally indicates poorer vehicle control. In critical cases, the vehicle may "weave," causing varying distances

from the centre. In order to ensure safe vehicle control, it is necessary to establish a stable driving path and reduce vehicle oscillation. Research has shown that variables such as phone calls, driver relaxation, and the watching of videos have been found to be significant contributors to increased variability in driving path, thereby indicating suboptimal takeover conditions. Another significant variable in the context of takeover quality is situational awareness. The findings of this study demonstrated a significant decline in situational awareness during periods of phone usage. The present study, therefore, in accordance with international research, focuses on the impact of different types of activities or their characteristics. However, in contrast to numerous foreign studies, the present experimental study is distinguished by its examination of six non-driving-related tasks of varying types and characteristics. In contrast to the majority of foreign studies, which have focused on two to three non-driving-related tasks (e.g. [14, 20, 33], only a minority of studies have examined six or more activities [5, 23]. The incorporation of a more extensive array of activities has yielded novel insights into the impact of diverse species and the disparities between them. An augmented array of tasks permitted greater variability in the presented scenarios. Furthermore, the study enabled the observation of the impact of fatigue on takeover performance. A comprehensive review of extant studies reveals that the present study is among a select few that have measured multiple activities on a truck simulator with a motion platform. The majority of extant studies have focused on passenger vehicle simulators. A notable exception is the study by Zhang et al. [10], which investigated the differences in taking over the wheel between driving with system monitoring, without system monitoring while working on a tablet, and in a situation with eyes closed. The results obtained from this study indicated that total takeover times were significantly longer in driving without system monitoring, and there was a high degree of variability in both the driver-not-monitoring and eyes-closed conditions, in comparison to the driver-monitoring condition. The results suggest that a personalised driver readiness predictor is an important input parameter for a safe and comfortable transition of control [10]. This finding is consistent with the results obtained in this study – when monitoring the system exclusively, the takeover time is shorter compared to a situation where drivers are engaged in other activities.

4.1 Limitations

The experiment was conducted on a Volvo truck simulator, which differs in terms of ergonomics and controls from future L3/L4 autonomous vehicles. The research sample consisted of experienced professionals. It is hypothesised that the results obtained from this

professional sample will differ from those obtained from non-professional drivers who have less experience and driving skills. The experimental design involved driving on a motorway and creating several situations for the driver to react to. Situations representing regular traffic were deliberately chosen, rather than collision or risk situations that are often simulated in other studies. However, this approach has the disadvantage of not observing erroneous responses in the form of early non-takeover and subsequent threatened or realised collisions (TTC, collisions, etc.), but only trends in takeover rates, and lateral and longitudinal changes.

4.2 Implications for the praxis

It is necessary to limit the performance of some risky types of activities, ideally through legislative measures. It is also advisable to inform drivers about the process of taking over control and the impact of different types of activities. Furthermore, it has been demonstrated that it is of paramount importance to provide drivers with comprehensive support throughout the process of taking over the wheel. This necessitates ensuring that drivers are well-informed about the process and can engage in it. The vehicles themselves will also play an important role, e.g., through well-designed dashboards and infotainment settings. In the context of driver education, the utilisation of checklists to support driver readiness for operating an autonomous vehicle or during the process of taking over control is a recommended practice.

5 Conclusions

Engagement in NDRTs has been demonstrated to increase takeover time, whilst concomitantly resulting in a deterioration of qualitative parameters. Activities requiring manual dexterity, such as those involving hand-held objects, result in prolonged takeover times when compared to activities that do not require such dexterity. Visual distraction has been identified as a significant factor in this regard. Mental workload has been shown to prolong takeover time and deteriorate other parameters. The present study has identified that for L3, the least appropriate tasks are those that are highly demanding, contain a visual modality, and include handholding. The majority of activities studied were inappropriate for L3. The audio course has been identified as the only viable option. A comprehensive analysis encompassing all measured parameters and respondents' perspectives reveals that working on a laptop exerts the most detrimental influence (visual activity, elevated mental demands, handholding). A similar negative effect was observed regarding watching a video, which, although it involves visual activity and medium mental demands, does not involve handholding. Another activity that has been shown to have a negative impact is making a phone

call (auditory activity, medium mental demands, no handholding). The influence of making a call was particularly evident in the qualitative aspects of the subsequent drive, while reaction time was, on the other hand, relatively short. However, respondents perceived this activity as relatively safe. Conversely, the manual activity (visual activity, low mental load, handholding) exhibited high latency, primarily attributable to object manipulation, resulting in prolonged reaction times. However, this activity also demonstrated a positive effect, with drivers being appropriately activated and ready to react. The relaxation activity (auditory activity, low mental load, no handholding) demonstrated both a negative and a positive influence. The positive effect was characterised by the subjects' subjective experience of relaxation or refreshment, which manifested as reduced reaction times. In terms of the qualitative aspects of driving, this activity shows a certain degree of riskiness due to the 'eyes closed' factor, which may cause a decline in situational awareness. Conversely, the least risky activity was found to be listening to an audio course, which is considered to be the least risky activity. This auditory activity demonstrated favourable outcomes across the majority of the parameters under study.

Abbreviations

CDV	Transport Research centre
L0	no automation
L1	vehicles with driver assistance
L2	partial driving automation
L3	conditional automation
L4	high automation
L5	full automation
M	mean
NDRT	non-driving-related task
TOR	takeover request
SAE	SAE International
SAGAT	Situational awareness global assessment technique
TTC	time-to-collision
SD	standard deviation
SDLP	the standard deviation of the vehicle's lateral position

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Author contributions

Miroslava Horáková was leading the research team on project „Systemization of non-driving activities in the autonomous mode“. She wrote the article with the data and outputs obtained through the project. Following authors contributed to the realization of the project and preparing the manuscript: Lucie Vondráčková processed the data using statistical analyses, contributed to the data description and interpretation. Libor Krejčí was responsible for methodology and data acquisition, contributed to the creation of the project's outputs. Martina Trepáčová processed the data using statistical analyses and contributed to the creation of the main project's outputs. All authors have read and agreed the final version of manuscript.

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Data availability

The datasets generated and analysed during the study are available in the Zenodo repository [<https://doi.org/10.5281/zenodo.15281989>].

Declarations

Competing interests

The authors declare that they have no competing interests.

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References

- Fagnant, D. J., & Kockelman, K. (2015). Preparing a nation for autonomous vehicles: Opportunities, barriers and policy recommendations. *Transportation Research Part A Policy and Practice*, 77, 167–181. <https://doi.org/10.1016/j.tri.2015.04.003>
- Nasr, V., Wozniak, D., Shahini, F., & Zahabi, M. (2021). Application of advanced driver-assistance systems in police vehicles. *Transportation Research Record: Journal of the Transportation Research Board*, 2675(10), 1453–1468. <https://doi.org/10.1177/03611981211017144>
- Litman, T. (2024). *Autonomous vehicle implementation predictions: Implications for transport planning*. Vitoria transport policy institute. Assessed 19 February 2025 <https://www.vtpi.org/avip.pdf>
- SAE International. (2021). Taxonomy and definitions for terms related to on-road motor vehicle automated driving systems. Accessed 12 August 2024 https://www.sae.org/standards/content/j3016_202104
- Lee, S. C., Yoon, S. H., & Ji, Y. G. (2020). Effects of non-driving-related task attributes on takeover quality in automated vehicles. *International Journal of Human-Computer Interaction*, 37. <https://doi.org/10.1080/10447318.2020.1815361>
- De Winter, J., Stanton, N. A., Price, J. S., & Mistry, H. (2016). The effects of driving with different levels of unreliable automation on self-reported workload and secondary task performance. *International Journal of Vehicle Design*, 70(4), 297–324. <https://doi.org/10.1016/j.trf.2014.06.016>
- Gold, G., Damböck, D., Lorenz, L., & Bengler, K. (2013). "Take over!" How long does it take to get the driver back into the loop? *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 57(1), 1938–1942. <https://doi.org/10.1177/0018720816634226>
- Shahini, F., & Zahabi, M. (2022). Effects of levels of automation and non-driving related tasks on driver performance and workload: A review of literature and meta-analysis. *Applied Ergonomics*, 104, 1–16. <https://doi.org/10.1016/j.apergo.2022.103824>
- Zhang, B., De Winter, J., Varotto, S., Happee, R., & Martens, M. (2019a). Determinants of take-over time from automated driving: A meta-analysis of 129 studies. *Transportation Research. Part F, Traffic Psychology and Behaviour*, 64, 285–307. <https://doi.org/10.1016/j.trf.2019.04.020>
- Zhang, B., Wilschut, E. S., Willemsen, D. M., & Martens, M. H. (2019). Transitions to manual control from highly automated driving in non-critical truck platooning scenarios. *Transportation Research Part F: Traffic Psychology and Behavior*, 64, 84–97. <https://doi.org/10.1016/j.trf.2019.04.006>
- Hungund, A. P., & Pradhan, A. K. (2023). Impact of non-driving related tasks while operating automated driving systems (ADS): A systematic review. *Accident Analysis and Prevention*, 188, 107076. <https://doi.org/10.1016/j.aap.2023.107076>
- De Winter, J., Happee, R., Martens, M., & Stanton, N. A. (2014). Effects of adaptive cruise control and highly automated driving on workload and situation awareness: A review of the empirical evidence. *Transportation Research Part F: Traffic Psychology and Behaviour*, 24, 196–217. <https://doi.org/10.1016/j.trf.2014.06.016>
- Müller, A. L., Fernandes-Estrela, N., Hetfleisch, R., Zecha, L., & Abendroth, B. (2021). Effects of non-driving related tasks on mental workload and take-over times during conditional automated driving. *European transport research review*, 13(1), 1–15. <https://doi.org/10.1186/s12544-021-00475-5>
- Zhang, N., Fard, M., Xu, J., Davy, J. L., & Robinson, S. R. (2023b). Influence of non-driving related tasks on driving performance after takeover transition in

- conditionally automated driving. *Transportation research part F: traffic psychology and behaviour*, 96, 248–264. <https://doi.org/10.1016/j.trf.2023.05.009>
15. Bourrelly, A., de Naurois, C., Zran, A., Rampillon, F., Vercher, J.-L., & Bourdin, C. (2019). Long automated driving phase affects take-over performance. *IET Intelligent Transport Systems*, 13, 1249–1255. <https://doi.org/10.1049/iet-its.2019.0018>
 16. Minhas, S., Hernández-Sabaté, A., Ehsan, S., & McDonald-Maier, K. D. (2022). Effects of non-driving related tasks during self-driving mode. *IEEE Transactions on Intelligent Transportation Systems*, 23(2), 1391–1399. <https://doi.org/10.1109/TITS.2020.3025542>
 17. Dogan, E., Honnet, V., Masfrand, S., & Guillaume, A. (2019). Effects of non-driving-related tasks on takeover performance in different takeover situations in conditionally automated driving. *Transportation Research. Part F, Traffic Psychology and Behaviour*, 62, 494–504. <https://doi.org/10.1016/j.trf.2019.02.010>
 18. Eriksson, A., & Stanton, N. A. (2017). Takeover time in highly automated vehicles: Noncritical transitions to and from manual control. *Human Factors*, 59(4), 689–705. <https://doi.org/10.1177/0018720816685832>
 19. Vogelpohl, T., Gehlmann, F., & Vollrath, M. (2019). Task interruption and control recovery strategies after take-over requests emphasize need for measures of situation awareness. *Human Factors: The Journal of the Human Factors & Ergonomics Society*, 62(7), 1190–1211. <https://doi.org/10.1177/0018720819866976>
 20. Yoon, S. H., & Ji, Y. G. (2019). Non-driving-related tasks, workload, and takeover performance in highly automated driving contexts. *Transportation Research. Part F, Traffic Psychology and Behaviour*, 60, 620–631. <https://doi.org/10.1016/j.trf.2018.11.015>
 21. Zeeb, K. (2016). *Der Einfluss fahrfremder Tätigkeiten auf die Fahrerübernahme während des hochautomatisierten Fahrens* [Dissertation. Heinrich-Heine-Universität]. Düsseldorf. Accessed 12 August 2024. <https://docserv.uni-duesseldorf.de/servlets/DocumentServlet?id=41406>
 22. Jin, L., Liu, X., Guo, B., Han, Z., Wang, Y., Cao, Y., Yang, X., & Shi, J. (2025). Impact of non-driving related task types, request modalities, and automation on driver takeover: A meta-analysis. In *Safety science* (Vol. 181). Elsevier B.V. <https://doi.org/10.1016/j.ssci.2024.106704>
 23. Müller, A. L. (2020). *Auswirkungen von naturalistischen fahrfremden Tätigkeiten während hochautomatisierter Fahrt*. Dissertation. Technische Universität, Darmstadt. 10.25534/tuprints-00011342. Accessed 25 August 2024.
 24. Naujoks, F., Höfling, S. P., C., & Zeeb, K. (2018). From partial and high automation to manual driving: Relationship between non-driving related tasks, drowsiness and take-over performance. *Accid. Anal. Prev.*, 121, 28–42. <https://doi.org/10.1016/j.aap.2018.08.018>
 25. Hu, W., Zhang, T., Zhang, Y., & Chan, A. H. S. (2024). Non-driving-related tasks influencing drivers' takeover time: A meta-analysis. *Transportation Research. Part F, Traffic Psychology and Behaviour*, 103, 623–637. <https://doi.org/10.1016/j.trf.2024.05.012>
 26. Wandtner, B., Schöming, N., & Schmidt, G. (2018). Effects of non-driving related task modalities on takeover performance in highly automated driving. *Human Factors*, 60(6), 870–881. <https://doi.org/10.1177/0018720818768199>
 27. Jeong, H., & Liu, Y. (2019). Effects of non-driving-related-task modality and road geometry on eye movements, lane-keeping performance, and workload while driving. *Transportation Research Part F: Traffic Psychology and Behaviour*, 60, 157–171. <https://doi.org/10.1016/j.trf.2018.10.015>
 28. Choi, D., Sato, T., Ando, T., Abe, T., Akamatsu, M., & Kitazaki, S. (2020). Effects of cognitive and visual loads on driving performance after take-over request (TOR) in automated driving. *Applied Ergonomics*, 85, 1–9. <https://doi.org/10.1016/j.apergo.2020.103074>
 29. Pan, H., Payre, W., Wang, Y., & Gao, Z. (2024). How do visual and cognitive non-driving related tasks affect drivers' visual attention and takeover performance in conditionally automated driving? *Journal of Transportation Safety & Security*, 17(1), 97–123. <https://doi.org/10.1080/19439962.2024.2368118>
 30. Guo, L., Xu, L., Ge, P., & Wang, X. (2023). How resource demands of Nondriving-related tasks and engagement Time affect Drivers' physiological response and takeover performance in conditional automated driving. *IEEE Transactions on Human-Machine Systems*, 53(3), 600–609. <https://doi.org/10.1109/THMS.2023.3268095>
 31. Zeeb, K., Haertel, M., Buchner, A., & Schrauf, M. (2017). Why is steering not the same as braking? The impact of non-driving related tasks on lateral and longitudinal driver interventions during conditionally automated driving. *Transportation Research. Part F, Traffic Psychology and Behaviour*, 50, 65–79. <https://doi.org/10.1016/j.trf.2017.07.008>
 32. Wan, J., & Wu, C. (2018). The effects of lead time of take-over request and nondriving tasks on taking-over control of automated vehicles. *IEEE Transactions on Human-Machine Systems*, 48(6), 582–591. <https://doi.org/10.1109/THMS.2018.2844251>
 33. Shi, E., & Bengler, K. (2022). Non-driving related tasks' effects on takeover and manual driving behavior in a real driving setting: A differentiation approach based on task switching and modality shifting. *Accident Analysis and Prevention*, 178, 106844. <https://doi.org/10.1016/j.aap.2022.106844>
 34. Yoon, S. H., & Lee, S. C. (2023). A systematic literature review on the effects of non-driving tasks on the takeover process in highly automated driving. *Ergonomics Society of Korea*, 42(1), 1–14. <https://doi.org/10.1016/j.aap.2018.11.018>
 35. Lu, Z., Coster, X., & de Winter, J. (2017). How much time do drivers need to obtain situation awareness? A laboratory-based study of automated driving. *Applied Ergonomics*, 60, 293–304. <https://doi.org/10.1016/j.apergo.2016.12.003>
 36. Strayer, D. L., & Fisher, D. L. (2016). SPIDER: A framework for understanding Driver distraction. *Human Factors: The Journal of the Human Factors & Ergonomics Society*, 58(1), 5–12. <https://doi.org/10.1177/0018720815619074>
 37. Rauch, N. (2009). *Ein verhaltensbasiertes Messmodell zur Erfassung von Situationsbewusstsein im Fahrkontext*. Inaugural-Dissertation zur Erfassung der Doktorwurde. Julius-Maximilians-Universität Würzburg.
 38. Laan, J. D., Heino, A., & Waard, D. (1997). A simple procedure for the assessment of acceptance of advanced transport telematics. *Transportation Research Part C*, 5(1), 1–10. [https://doi.org/10.1016/S0968-090X\(96\)00025-3](https://doi.org/10.1016/S0968-090X(96)00025-3)
 39. Körber, M., Prasch, L., & Bengler, K. (2018). Why Do I Have to Drive Now? Post Hoc Explanations of Takeover Requests. *Human Factors*, 60(3), 305–323. <https://doi.org/10.1177/0018720817747730>
 40. Zhang, N., Fard, M., Davy, J. L., Parida, S., & Robinson, S. R. (2023). Is driving experience all that matters? drivers' takeover performance in conditionally automated driving. *Journal of Safety Research*, 87, 323–331. <https://doi.org/10.1016/j.jsr.2023.08.003>
 41. Petermann-Stock, I., Hackenberg, L., Muhr, T., & Mergl, C. (2013). Wie lange braucht der Fahrer? Eine Analyse zu Übernahmezeiten aus verschiedenen Nebentätigkeiten ährend einer hochautomatisierten Staufahrt. In S. Tüv (Ed.), *Der Weg zum automatischen Fahren. 6. Tagung Fahrerassistenz* (pp. 1–26).
 42. Soares, S., Lobo, A., Ferreira, S., Cunha, L., & Couto, A. (2021). Takeover performance evaluation using driving simulation: A systematic review and meta-analysis. *European Transport Research Review*, 13(1), 1–18. <https://doi.org/10.1186/s12544-021-00505-2>

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